SIGNAL & STRESS

From unstructured financial news to transparent portfolio stress scenarios.



AI / NLP RISK ENGINE

+ MODULE B: STRATEGIC PORTFOLIO STRESS TESTING

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12,000
synthetic text records

2,400
fictional holdings

2
live RSS publishers

The decision gap

The hard part is not finding a headline. It is translating it into an exposure decision.

ILLUSTRATIVE SCENARIO · FICTIONAL COMPANY

“Trade sanctions threaten critical supply routes.”

One headline. Three analyst questions.

01 What kind of event is this?

02 How adverse and how material?

03 Which exposures should I inspect?

SOURCE · SYNTHETIC DEMO EXAMPLE

FROM READING TO REVIEW

01 STRUCTURE
Classify as Geopolitical

02 PRIORITIZE
Estimate sentiment + impact

03 INVESTIGATE
Apply a defined portfolio scenario

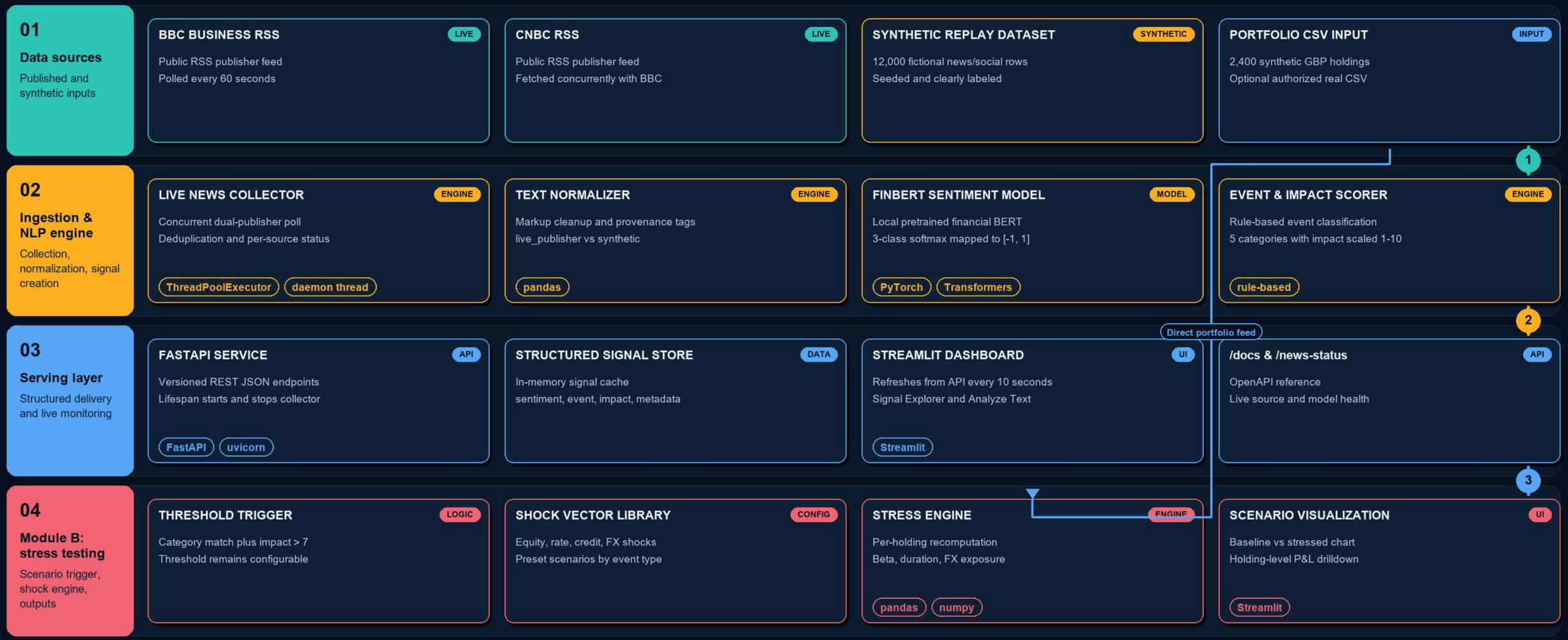
Decision support — not automated investment advice.

DESIGN PRINCIPLE

Keep source provenance, model output and scenario assumptions visible at every step.

SIGNAL & STRESS | SYSTEM ARCHITECTURE

End-to-end data flow: real and labeled-synthetic ingestion, local FinBERT scoring, structured API delivery, and event-driven portfolio stress testing



FLOW

1. Ingestion normalizes real and synthetic text and preserves provenance.
2. FinBERT and rules convert text into sentiment, event, and impact signals.
3. Signals are exposed via JSON and can trigger stress testing when severity is high.

LEGEND

- Real published data
- Synthetic or rule-based processing
- Computed or served signal
- Stress-testing application output

RSS polling is not publisher push streaming. Synthetic data remains labeled at schema level. Scenario losses are illustrative, not forecasts.

How a headline becomes a signal

Four explicit stages produce an auditable signal contract — with model and rule responsibilities separated.

INGEST & NORMALIZE

01

BBC + CNBC RSS

Parallel 60-second polling

Dedup + provenance tags

FINBERT SENTIMENT

02

Local ProsusAI FinBERT

Positive / negative / neutral

Signed confidence in [-1, 1]

EVENT CLASSIFICATION

03

Keyword-based event rules

Five named event classes

General Market fallback

IMPACT ESTIMATE

04

Category base score

+ round(2 x |sentiment|)

Clamped to [1, 10]

API SIGNAL CONTRACT

SENTIMENT	EVENT	IMPACT	TRACEABILITY
-1.0 to +1.0	category label	1 to 10	source · time · provenance

Honest boundary: event classification and impact are heuristic rules; UI highlights are keyword cues, not FinBERT explanations.

Module B | From event to portfolio scenario

A geopolitical example using a fictional wholesale-style portfolio and a predefined shock vector.

GEOPOLITICAL SHOCK ASSUMPTIONS

Equities	-10.0%
Interest rates	+2.0 pp
Credit spreads	+1.5 pp
FX factor	-5.0%

Trigger example: impact > 7 | dashboard threshold = 8

BUNDLED SYNTHETIC DATASET RESULT

BASELINE

GBP 14.69bn

STRESSED VALUE

GBP 13.39bn

-8.80%

-GBP 1.29bn

2,400 holdings · 240 fictional issuers · GBP

TRANSPARENT FIRST-ORDER SENSITIVITIES

EQUITY

value x beta x equity return

RATES

-value x duration x rate change

CREDIT

-value x credit duration x spread change

FX

currency exposure x FX return

Estimate only: linear sensitivities, zero floor and simplified derivatives; not a calibrated forecast or valuation.

Evidence, data and boundaries

A reproducible demo with public publisher inputs and explicitly fictional exposure data.

2
publisher RSS feeds

12,000
synthetic text records

2,400
synthetic holdings

37
tests passed at last run

DATA PROVENANCE

REAL

BBC Business + CNBC public RSS; collection metadata retained; polled every 60 seconds.

SYNTHETIC

Seed 20261008. Fictional entities, news/social text, GBP holdings, and bounded sensitivities. No confidential client data.

No live Twitter/X integration or market-price stream.

WHAT VERIFICATION MEANS

VERIFIED	Publisher collection, local inference path, API behavior, scenario math and tests.
NOT CLAIMED	Held-out FinBERT accuracy, calibrated impact prediction or measured analyst time saved.
REPRODUCIBILITY	Bundled CSV/JSONL, manifest, fixed seed and documented local run command.

DOMAIN CAVEAT RSS polling is not push streaming. Synthetic scenario P&L is illustrative, not a forecast.

A traceable path from news to risk review

Designed to support analyst triage and consistent what-if exploration — not replace expert judgment.

THE REVIEWER WALKTHROUGH

01 **Confirm live source + FinBERT status**

02 **Analyze the fictional Debt default case**

03 **Set threshold 8; test Trade sanctions**

04 **Inspect portfolio change + holding drivers**

05 **Open API docs and synthetic-data manifest**

NEXT ITERATIONS

- Evaluate on unseen headlines
- Calibrate severity with expert labels
- Improve entity-to-exposure mapping
- Extend derivative valuation

AI assistance used with review and disclosure.

TRACEABILITY

HEADLINE → SIGNAL → SCENARIO → PORTFOLIO IMPACT

Public repo: github.com/shaaran-r/vit-sharan-r-hackathon

Demo video: pending